



Basics & Power Semiconductor Devices

Thyristor:

The name 'thyristor' is derived by a Combination of the capital letters from THYRatron and transISTOR. International Electrotechnical Commission (IEC) in 1963 decided the definition of Thyristor as under:

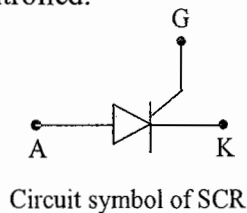
- (i) It constitutes three or more p - n junctions.
- (ii) It has two stable states, an ON - State and an OFF - State and can change its state from one to another. Thyristor is a four layer, three - junction, p - n - p - n semiconductor switching device.

Silicon Controlled Rectifier (SCR).

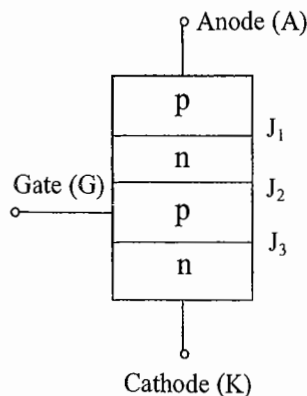
SCR is the oldest & first member of the thyristor family. It is called SCR because; silicon is used for its construction and its operation as a rectifier (very low resistance in forward conduction and very high resistance in the reverse direction) and can be controlled.

It has three terminals;

1. Anode (A)
2. Cathode (K)
3. Gate (G)



The structure of SCR



The terminal connected to outer 'p' region is called Anode (A)

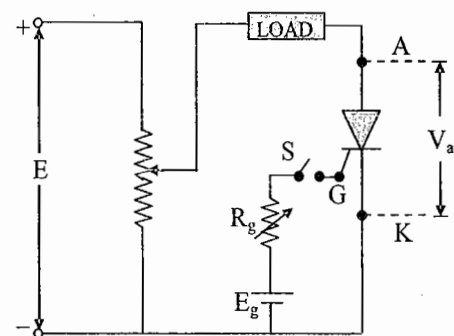
The terminal connected to outer 'n' region is called Cathode (K)

The terminal connected to inner 'p' region is called the Gate (G)

SCR is a unidirectional device. It blocks the current flow from cathode to anode.

Static V - I Characteristics of SCR

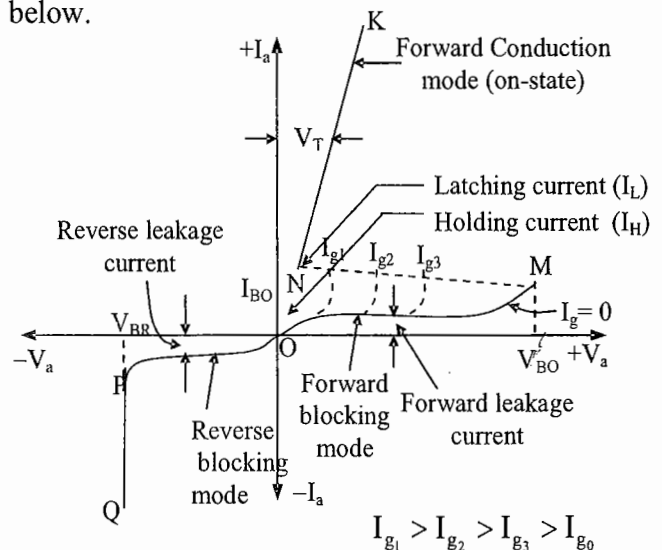
An elementary circuit diagram for obtaining static V - I characteristics of SCR is shown in Fig.



The anode and cathode are connected to main source through the load.

The Gate and cathode are fed from another source 'Eg'.

The static V - I characteristics of SCR are shown below.



V_a = Anode voltage; I_a = Anode current.

V_{BO} = Forward break over Voltage.

V_{BR} = Reverse breakdown voltage.

I_g = Gate current.



I_L = Latching current.

I_H = Holding current.

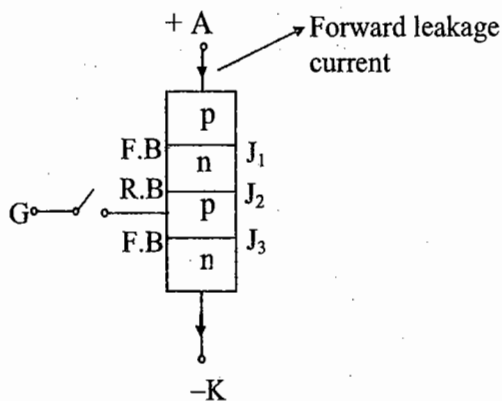
The three basic modes of operation of SCR are:

1. Forward Blocking mode.
2. Forward Conduction mode.
3. Reverse Blocking mode.

1. Forward Blocking Mode:

- When anode is at a higher potential than cathode, thyristor is said to be forward biased.

It is seen from the figure that when gate circuit is open J_1, J_3 are forward bias but junction J_2 is reverse bias.



- In this mode, a small current, called forward leakage current flows from anode to cathode.
- OM in V-I characteristics represents the forward blocking mode of SCR.
- SCR is treated as an open switch in the forward blocking mode.

2. Forward Conduction Mode:

When anode to cathode forward voltage is increased with gate circuit open, reverse biased junction J_2 will have an avalanche breakdown at a voltage called forward break over voltage V_{BO} .

- After this breakdown, thyristor gets turned ON with point 'M' at once shifting to 'N'. Here NK represents the Forward conduction mode.

A thyristor can be brought from forward blocking mode to forward conducting mode by applying,

1. A positive gate pulse between gate and cathode.
- (or)
2. A forward break over voltage (V_{BO}) across anode and cathode.

Voltage drop across the SCR ' V_T ' increases slightly with an increase in anode current. It can be seen from NK.

LATCHING CURRENT (I_L):

It is defined as the minimum value of anode current (I_a) which it must attain during turn – on process to maintain conduction when gate signal is removed.

The gate pulse width should be chosen to ensure that the anode current rises above the latching current (I_L).

HOLDING CURRENT (I_H):

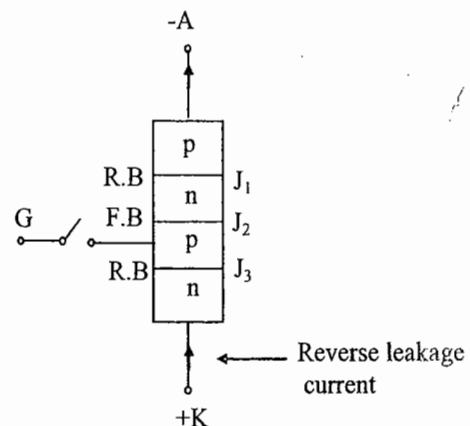
It is defined as the minimum value of anode current below which the SCR gets turned OFF.

Latching current (I_L) is more than holding current I_H .

Latching current (I_L) is associated with turn on process.

Holding current (I_H) is associated with turn off process.

3. REVERSE BLOCKING MODE:





Spread time (t_p):

The time taken by the anode current to rise from $0.9 I_a$ to I_a

(or)

It is also defined as the time for the forward blocking voltage to fall from 0.1 of its initial value to the on – state voltage drop.

Switching Characteristics during Turn – off:

SCR turn off means that it has changed from ON to OFF state and is capable of blocking the forward voltage. This dynamic process of the SCR from conduction state to forward blocking state is called commutation process or turn – off process.

If forward voltage is applied to the SCR at the moment its anode current falls to zero, the device will not be able to block this forward voltage, as the carriers (holes and electrons) in the four layers are still favorable for conduction. The device will therefore go into conduction immediately even though gate signal is not applied. To solve this problem it is essential that the thyristor is reverse biased for a finite period after the anode current has reached zero.

Thyristor Turn off time (t_q):

It is the time between the instant anode current becomes zero and the instant SCR regains forward blocking capability.

During this time (t_q) all the excess carriers from four layers of SCR are removed.

The turn off time is divided into two intervals.

- (a) reverse recovery time (t_{rr})
- (b) gate recovery time (t_{gr})

$$t_q = t_{rr} + t_{gr}$$

After ' t_1 ', anode current builds up in the reverse direction with the same $\frac{di}{dt}$ slope. The reason for

the reversal of anode current is due to the presence of charge carriers stored in the four layers.

At instant t_3 , when reverse recovery current has fallers to nearly zero value, end junctions J_1 and J_3 recover and SCR is able to block the reverse voltage.

at the end of reverse recovery period t_3 , the middle junction ' J_2 ' still has charges, therefore, the

thyristor is not able to block the forward voltage at t_3 .

The charge carriers at J_2 cannot flow to the external circuit therefore they must decay only by recombination. This is possible if a reverse voltage is maintained across SCR. The time taken for this ($t_4 - t_3$) is called gate recovery time (t_{gr}).

The thyristor turn off time ' t_q ' is depended up on magnitude of forward current, $\frac{di}{dt}$ at the time of commutation and junction temperature.

Circuit turn off time ' t_c ':

It is defined as the time between the instant anode current becomes zero and the instant reverse voltage due to practical circuit reaches zero.

$t_c > t_q$ for reliable turn off, otherwise the device may turn – on at an undesired instant, a process called Commutation failure.

Thyristors with slow turn – off time are called converter grade SCR's.

Ex: phase controlled rectifiers, cyclo converters & ac voltage controllers

SCR's with fast turn off time are called inverter grade SCR

Ex: inverters, choppers and forced commutation converters.



- 01. A power semi-conductor diode can safely carry the rated current from the anode to the cathode. But if it carries the same current in the reverse direction, it will be destroyed. The reason is**
- (a) The device has a reverse breakdown before it carries any current in the reverse direction.
 - (b) The device is not designed to carry current in the reverse direction.
 - (c) The voltage drop across the diode, when carrying current in the forward direction, is larger than it is when carrying current in the reverse direction.
 - (d) The voltage drop across the diode when carrying current in the reverse direction is larger than it is when carrying current in the forward direction.

Sol:



$V_2 \gg V_1$. Power dissipation (as heat) $V_2 I \gg V_1 I$

- 02.** The reverse recovery time of a diode is $3\mu\text{s}$ and the rate of fall of the diode current is $3\text{A}/\mu\text{s}$. The peak value of reverse current of the diode is given by

- (a) 135 A (b) 90 A
(c) 100 A (d) 8100 A

Sol: $Q_{RR} = \frac{1}{2} \frac{di}{dt} \times t_{rr}^2 = 0.5 \times 30 \times (3\mu)^2 = 135 \mu\text{C}$

$$I_{RR} = \left[2Q_{RR} \frac{di}{dt} \right]^{\frac{1}{2}} = 90 \text{ A}$$

GATE CHARACTERISTICS OF SCR:

Trigger circuit connected to gate – cathode circuit of an SCR

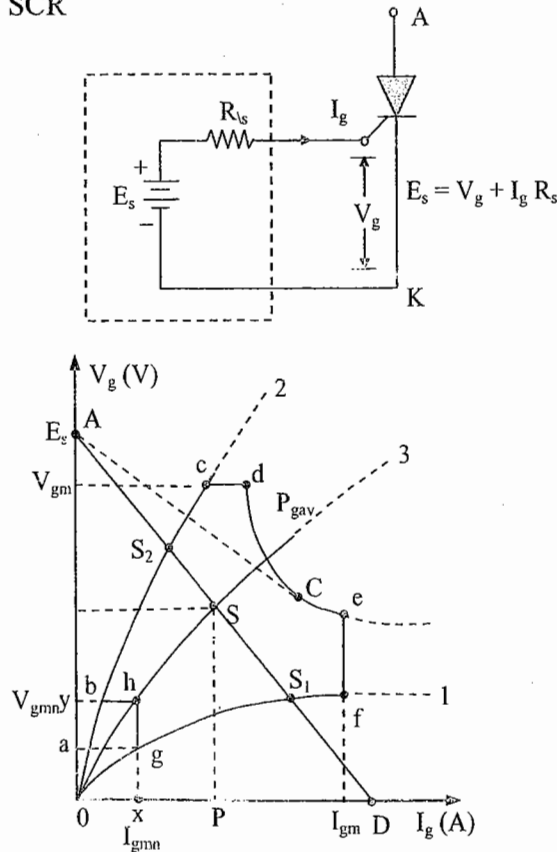


Fig. Forward Gate characteristics of SCR

V_g , I_g characteristics has spread between two curves (1) and (2) as shown in fig.

Curve (1) corresponds to gate with maximum doping. So it requires lowest voltage values that must be applied to turn – on the SCR.

Curve (2) corresponds to gate with minimum doping. So it requires highest possible voltage values that can be safely applied to gate circuit.

Each SCR has maximum limits as V_{gm} for gate voltage and I_{gm} for gate current. There is also average gate power dissipation ' P_{gav} ' specified for each SCR.

V_{gmn} and I_{gmn} are the minimum limits for gate voltage & gate current to turn on SCR.

The above figure shows that the preferred gate drive area for an SCR is 'bcdefghb'

For selecting the operating point of the circuit Load line is drawn from A to D. ($\overline{AD}$ = load line) where as.

E_s = gate source voltage = OA

$$OD = \text{Trigger circuit short circuit current} = \left[\frac{E_s}{R_s} \right]$$

Curve (3) gives the operating points 'S'.

Gate voltage = PS. and Gate current = OP.

Operating point 'S' which may change from S_1 to S_2 and must lie within the limits curves 1 and 2. and must be as close to the P_{gav} curve as possible.

The minimum value of gate source series resistance ' R_s ', is obtained by drawing a line AC tangent to P_{gav} curve.

Workedout Examples

- 03.** A thyristor data sheet gives 1.5 V and 100 mA as the gate-trigger voltage and gate-trigger current respectively. A resistance of 20Ω is connected across gate-cathode terminals. For a trigger supply voltage of 8 V, compute the value of resistance that should be connected in series with gate circuit in order to ensure turn on of the device

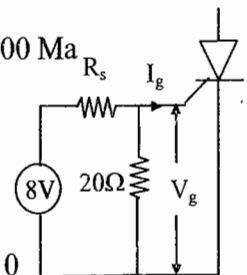
Sol: Given $V_g = 1.5 \text{ V}$, $I_g = 100 \text{ Ma}$

From circuit,

$$\frac{V_g - 8}{R_s} + \frac{V_g}{20} + I_g = 0$$

$$\frac{1.5 - 8}{R_s} + \frac{1.5}{20} + 100 \times 10^{-3} = 0$$

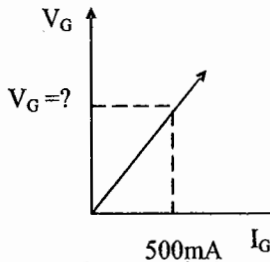
$$R_s = 37.14 \Omega$$





04. For an SCR, the gate cathode characteristics are given by a straight line passing through the origin with a gradient of 16V/A. The turn-on time is 4μs and the gate current required is 500mA. For the gate source voltage of 15V, gate power dissipation is _____

Sol:



From the gate characteristics

$$V_G = 16I_G$$

For 500mA, $V_G = 16 \times 0.5 \Rightarrow 8V$

$$\begin{aligned} \text{Therefore the gate power dissipation} &= V_G I_G \\ &= 8 \times 0.5 \\ &= 4W \end{aligned}$$

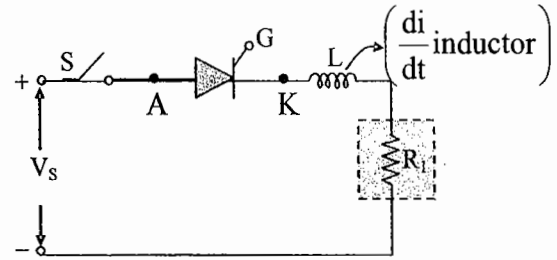
05. With a gate-cathode voltage constant at 5V, the gate current of a thyristor is observed to be 0.1 A. If gate pulses of duty cycle 0.7 are used, the average gate power dissipation (in watts) is
(a) 0.5 (b) 0.35 (c) 5 (d) 0.035

Sol: Average dissipation in one period = Average dissipation = $5 \times 0.1 \times 0.7 = 0.35 W$

PROTECTION OF SCR:

$\frac{di}{dt}$ Protection:

When a thyristor is forward biased and is turned ON by a gate pulse, conduction of anode current begins. If the rate of rise of anode current $\left(\frac{di}{dt}\right)$ is very fast compared with the spreading velocity of charges a local hot spots will be formed. This localised heating may destroy the thyristor. Therefore, the $\left(\frac{di}{dt}\right)$ rate of rise of anode current at the time of turn on must be kept below the specified limiting value by using a small inductor, called $\frac{di}{dt}$ inductor, in series with the anode circuit.



$\frac{dv}{dt}$ Protection:

If the rate of rise of applied voltage across thyristor is high, the device may get turned on. Such phenomena of turning on a thyristor, called $\frac{dv}{dt}$ turn on must be avoided as it leads to false operation.

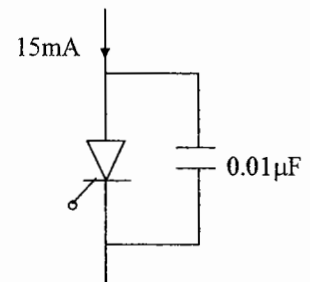
False turn on of an SCR by large $\frac{dv}{dt}$, even without application of gate signal can be prevented by using a Snubber circuit in parallel with the device.

Workedout Examples

06. The junction capacitance of a thyristor is $C_j = 20 \text{ pF}$ and can be assumed independent of off-state voltage. The limiting value of charging current to turn on the thyristors is 15 mA. If a capacitor of $0.01 \mu\text{F}$ is connected across the thyristor, determine the critical value of $\frac{dv}{dt}$.

Sol: $\frac{dv}{dt} = ?$

$$I_C = C_{\text{equ}} \frac{dv}{dt}$$



$$15 \times 10^{-3} = (20 \times 10^{-12} + 0.01 \times 10^{-6}) \frac{dv}{dt}$$

$$\frac{dv}{dt} = 1.5 \text{ V}/\mu\text{sec}$$

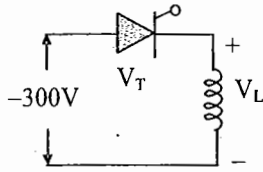
07. During the turn-off process in a thyristor, the reverse recovery current of 10A is interrupted in a time interval of 4μs. The thyristor is connected in series with an inductance of 6mH with no resistance in the circuit. If the source



voltage during turn-off process is -300V , peak voltage across the thyristor when reverse current interrupted is

- (a) -10 kV (b) -18.7 kV
(c) -12.7 kV (d) -15.3 kV

Sol:



The voltage across the inductor during this period is $V_L = L \frac{di}{dt}$

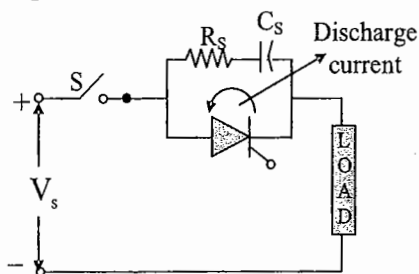
$$V_L = 6 \times 10^{-3} \left(\frac{10}{4 \times 10^{-6}} \right) = 15\text{ kV}$$

Peak voltage across the thyristor is
 $300 + V_T + 15000 = 0$
 $V_T = -15.3\text{ kV}$

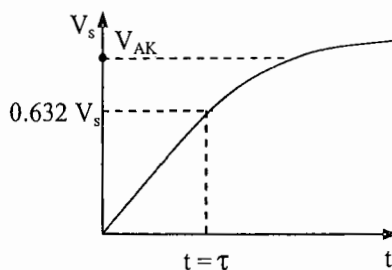
Design of snubber circuit:

A snubber circuit consists of a series combination of resistance R_s and capacitance C_s in parallel with the thyristor as shown in fig.

When switch 'S' is closed sudden voltage is appeared across the capacitor C_s and the thyristor is open circuited. Voltage across the capacitor C_s builds up at a slow rate.



Snubber circuit across SCR



The $\frac{dv}{dt}$ across ' C_s ' is low and therefore across

SCR is less than the specified maximum $\frac{dv}{dt}$ rating of the device.

In order to limit the magnitude of discharge current, a resistance R_s is inserted in series with C_s .

Over voltage protection:

An SCR may be subjected to internal or external over voltages.

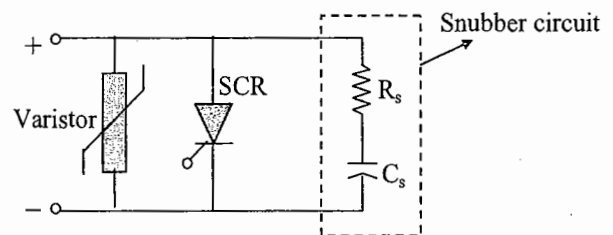
Internal over voltages:

Due to the presence of the series inductance L_s of the SCR circuit, large transient voltages ($L \frac{di}{dt} = V_L$) is produced and this internal voltage may be several times the break over voltage of the SCR, the thyristor may be destroyed permanently.

External over voltages:

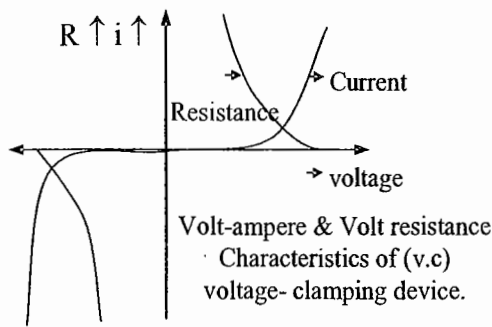
Over voltages are caused due to the lightning strokes and switching surges.

For reliable operation, the over voltages must be suppressed by adapting voltage – clamping (V.C) device.



A voltage – clamping (V.C) device is a non – linear resistor connect across SCR as shown in fig. The V.C. device has falling resistance characteristics with increasing voltage. Under normal conditions the device has a high resistance and draws only a small leakage current.

When a voltage surge appears, the V.C. device operates in Low resistance region and produces a short circuit across SCR. After the surge is dissipated it returns to normal high resistance.



Over - Current Protection:

If an SCR is subjected to an over current due to faults, short circuit (or) surge currents, its junction temperature may exceed the rated value and the device may be damaged. There is a need for the over current protection of SCR.

- (i) Fast Acting Current Limiting Fuse (FACLF).
- (ii) Circuit Breakers.

GATE Protection:

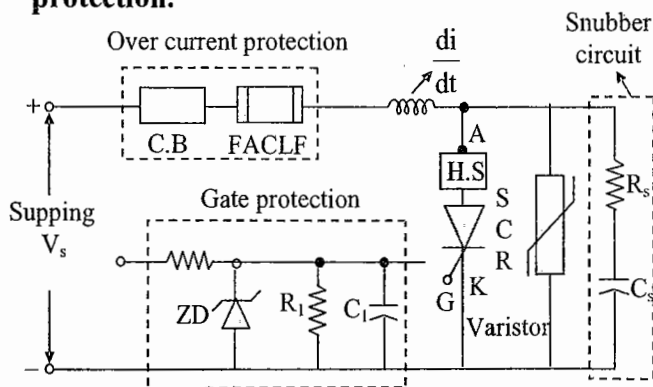
Over voltages and over currents across the gate circuit can cause false triggering of the SCR.

Protection against over - voltages is achieved by connecting a zener diode ZD across the gate circuit.

Protection against the over - current is achieved by connecting a resistor 'R₂' in series with gate circuit.

A capacitor and a resistor are also connected across gate to cathode to by pass the noise signals.

Circuit components showing the thyristor protection.



Thermal protection:

If temperature increases, insulation is weakened, resulting in leakage currents which interfere with

the operation. Heat sinks which include aluminum discs are used along with ventilating ducts, increased surface area, coolants are used.

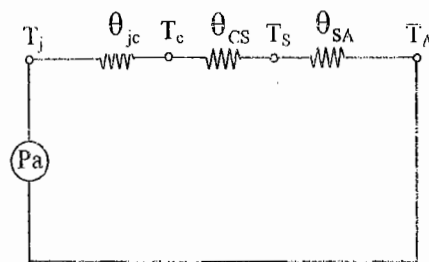
Heating and cooling of SCR:

The rate of heat transfer (i.e) current flow is constant.

Thermal resistance is denoted by ' θ '. If power loss P_{av} in watts causes the temperature of two points to be at $T_1^{\circ}\text{C}$ and $T_2^{\circ}\text{C}$ where $T_1 > T_2$ then thermal resistance is given by

$$\theta_{12} = \frac{T_1 - T_2}{P_{av}} \text{ }^{\circ}\text{C/W}$$

Various temperature and thermal resistances are shown in the below fig.



The thermal resistance between the junction temperature T_j & case temperature T_c is ' θ_{jc} '.

The thermal resistance between the case temperature T_c & Sink temperature T_s is ' θ_{cs} '. The thermal resistance between the sink temperature T_s & ambient temperature T_a is ' θ_{sa} '. P_{av} is the average rate of heat generated at a thyristor junction and it is electrical analogous to constant - current source Here.

$$P_{av} = \frac{T_j - T_c}{\theta_{jc}} = \frac{T_c - T_s}{\theta_{cs}} = \frac{T_s - T_a}{\theta_{sa}}$$

$$P_{av} = \frac{T_j - T_a}{\theta_{ja}}$$

Where, $\theta_{ja} = \theta_{jc} + \theta_{cs} + \theta_{sa}$ is the total thermal resistance between junction and ambient.

The difference in temperature between junction and ambient can be written in

$$T_j - T_a = P_{av} (\theta_{ja})$$

This shows that for maximum value of T_j , P_{av} can be increased by reducing ' θ_{ja} '. This means that by providing efficient cooling system to the SCR, the



power dissipation capability of the device can be increased.

Workedout Examples

08. The maximum junction temperature of a transistor is 150°C and the ambient temperature is 25°C . If the total thermal impedance is 1°C/W . What is the maximum power dissipation?

- (a) $\frac{1}{175}$ W (b) 175 W
(c) 125 W (d) $\frac{1}{125}$ W

Sol:

$$P_{av} = \frac{T_j - T_A}{\theta_{jA}} = \frac{150 - 25}{1} = 125\text{W}$$

09. A thyristor converter of 415V, 100A is operating at rated load. Details of the thyristor used are as follows. 'ON' state power loss = 150W Thermal

resistance: Junction to case = 0.01°C/W

Case to sink = 0.08°C/W Sink to atmosphere = 0.09°C/W .

Assume ambient temperature as 35°C . The junction temperature for 100% load is

- (a) 48.5°C (b) 54.5°C
(c) 60°C (d) 62°C

$$P_{av} = \frac{T_j - T_A}{\theta_{jA}} \Rightarrow 150 = \frac{T_j - 35}{0.18}$$

$$T_j = 62^{\circ}\text{C}$$

SERIES AND PARALLEL OPERATION OF SCR:

For industrial applications, the demand for voltage and current rating is so high that a single SCR cannot fulfill such requirement. In such cases SCR's are connected.

- (a) In series in order to meet the high voltage demand.
(b) In parallel for fulfilling the high current demand.

In series or parallel connected SCRs, it should be ensured that each SCR rating is fully utilized and the system operation is satisfactory. String

efficiency is a term that is used for measuring the degree of utilization of SCRs in a string.

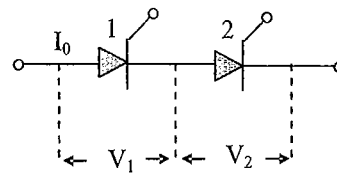
String efficiency

$$= \frac{\text{Actual voltage / current rating of the whole string}}{(\text{individual voltage / current rating of one SCR})} \times (\text{Number of SCR}_s \text{ in the string})$$

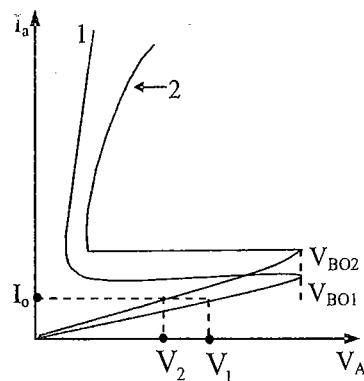
A measure of reliability of string is given by a factor called Derating factor (DRF). It is defined as:

$$\text{DRF} = 1 - \text{String efficiency.}$$

SERIES OPERATION:



To connect SCR's in series their I - V characteristics should be as close as possible.



$$\text{The string efficiency, } \eta_s = \frac{V_1 + V_2}{2V_1}$$

A more practical way of obtaining a reasonable uniform voltage distribution during steady state working of series connected SCRs is to connect the same value of shunt resistance R across each SCR. The shunt resistance 'R' is called the **Static Equalizing Resistor**.

Workedout Example

10. A certain SCR is rated 500V, 100A. What is the number of series SCR'S required for a circuit of 10 kV, 100A. If the derating factor is 15%.

(a) 17 (b) 20 (c) 24 (d) 30

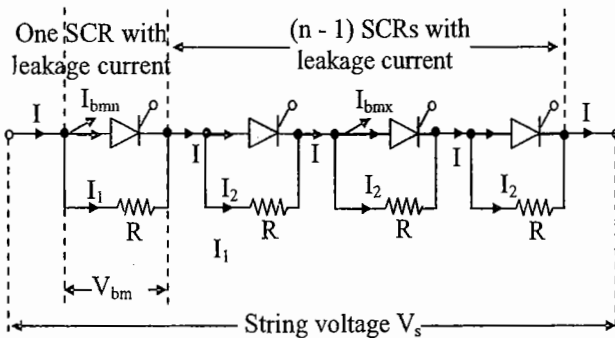
Sol: String efficiency = $\frac{10 \times 10^3}{500 \times n}$

DRF = 1 - string efficiency

$0.15 = 1 - \frac{10 \times 10^3}{500 \times n}$

$N = 23.52 \approx 24$

Circuit:



Consider n thyristors connected in series as shown in the fig.

Let SCR₁ has minimum leakage current I_{bmn} . SCR with lower leakage current blocks more voltage.

Remaining $(n-1)$ SCRs have the same leakage current I_{bmx} and $I_{bmx} > I_{bmn}$.

Here V_{bm} is the maximum permissible blocking voltage of SCR₁.

$$I = I_1 + I_{bmn} ; I = I_2 + I_{bmx}$$

$$I_1 = I - I_{bmn} ; I_2 = I - I_{bmx}$$

Where I = total string current.

Voltage across SCR₁ is $V_{bm} = I_1 R$

Voltage across $(n-1)$ SCR_s = $(n-1) I_2 R$

For a string voltage (V_s), the voltage equation for the series circuit is

$$\begin{aligned} V_s &= I_1 R + (n-1) I_2 R \\ &= V_{bm} + (n-1) R (I - I_{bmx}) \\ &= V_{bm} + (n-1) R (I_1 + I_{bmn} - I_{bmx}) \\ &= V_{bm} + (n-1) R [I_1 - (I_{bmx} - I_{bmn})] \end{aligned}$$

$$V_s = V_{bm} + (n-1) R I_1 - (n-1) R \Delta I_b$$

$$\because [I_b = I_{bmx} - I_{bmn}]$$

$$\text{As } R I_1 = V_{bm},$$

$$V_s = V_{bm} + (n-1) V_{bm} - (n-1) R \Delta I_b$$

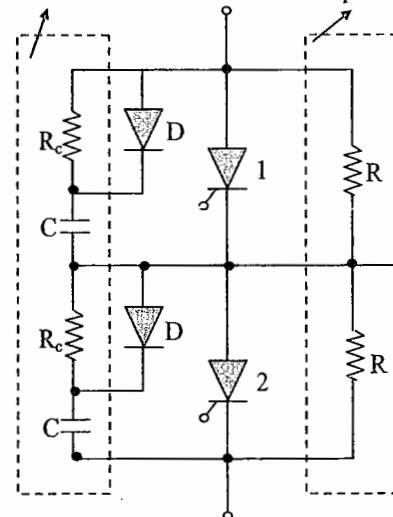
$$V_s = n V_{bm} - (n-1) R \Delta I_b$$

$$R_s = \frac{n V_{bm} - V_s}{(n-1) \Delta I_b} \Omega$$

' R_s ' is the static equalizing resistance.

Dynamic and static equalizing circuit for series connected SCR's.

Dynamic equalizing circuit Static equalizing circuit



A simple resistor as shown in static voltage equalization cannot maintain equal voltage distribution under transient condition. The shunt capacitors play a dominant role in equalizing the voltage distribution across the series connected thyristor during turn on and turn - off process.

When any SCR in the forward blocking state, the capacitor connected across it gets charged to a voltage existing across that SCR. For limiting this discharge current Damping resistor R_c is used in series with capacitor C .

The resistor R_c also damps out the high frequency oscillation that may arise due to series combination of R_c , shunt capacitor and circuit inductance.

Combination of ' R_c ' and ' C ' is called Dynamic Equalizing Circuit.

A diode D is also placed across ' R_c '. When forward voltage appears, diode bypasses through R_c during charging time of the capacitor C .

This makes the capacitor more effective in voltage equalization and for limiting $\frac{dv}{dt}$ across SCR.

However, during capacitor discharge, R_C comes in to play for limiting the current.

Value of the capacitance 'C' can be obtain as under:

In series connected SCR's, voltage unbalance during turn off time is more predominant than it is during turn on time, therefore choice of capacitor 'C' is based on the reverse recovery characteristics of SCR.

During turn off process, the source voltage V_s must reverse to aid the reverse recovery current.

The transient voltage which each SCR must be able to withstand is ' V_{bm} '.

Total voltage acting across the circuit consisting of V_s , thyristors $n, n-1, \dots, 3, 2$ and top 'C'. as per

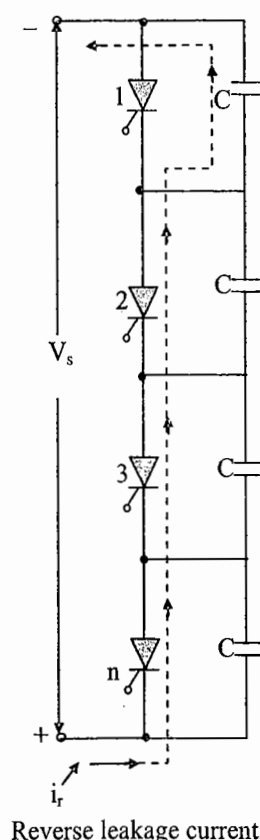
KVL = $V_s + \frac{(n-1)\Delta Q}{C}$ and must be supported by

all SCRs together, which is equal to $n \cdot V_{bm}$. $\frac{\Delta Q}{C}$ is the voltage induced by ΔQ in the capacitor C.

$$\text{So, } n V_{bm} = V_s + \frac{(n-1)\Delta Q}{C}$$

$$V_{bn} = \frac{1}{n} \left[V_s + \frac{(n-1)\Delta Q}{C} \right]$$

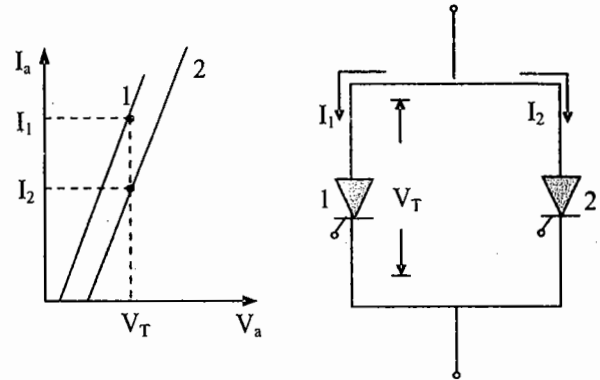
$$C = \frac{(n-1)\Delta Q}{n \cdot V_{bm} - V_s} \text{ F}$$



Parallel Operation:

When current required by the load is more than the rated current of a single thyristor, SCRs are connected in parallel in a string.

For equal sharing of currents, $I-V$ characteristics of SCRs during forward conduction must be identical as far as possible.



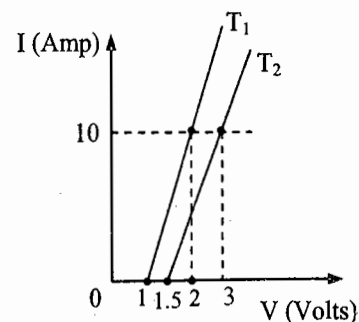
$$\text{String efficiency } \eta_s = \frac{I_1 + I_2}{I_1 \times 2}$$

$$\eta_s = \frac{I_1 + I_2}{2I_1}$$

The unequal current distribution and other operating problems are discussed below.

Workedout Example

11. Two thyristors T_1 and T_2 are being operated in parallel. Their voltage-current characteristic when they are ON are given in figure. The common voltage across T_1 and T_2 is 1.8 V. The heat dissipation in T_1 and T_2 respectively (in watts) is



- (A) 8, 2
(C) 20, 30

- (B) 64, 4
(D) 14.4, 3.6

Sol: The voltage-current equations related to T_1 and T_2 are $I = 10V - 10$; and $I = \frac{20}{3}V - 10$

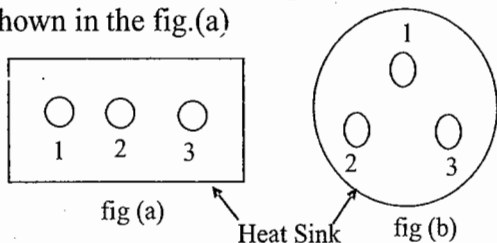
respectively. With a common voltage of 1.8V across the parallel combination of T_1 and T_2 $I_{T1} = 8$ A and $I_{T2} = 2$ A. The power dissipations are $8 \times 1.8 = 14.4$ W and $2 \times 1.8 = 3.6$ W respectively.

(i) **Simultaneous turn – on:** consider ‘n’ parallel connected SCRs. For satisfactory operation of these SCRs, they should get turned on at the same moment.

(ii) **Same temperature rise:** when SCRs are to be operated in parallel, It should be ensured that they operate at the same temperature. This can be achieved by mounting the parallel unit on one common heat sink.

(iii) **Symmetrical arrangement:**

When SCRs are arranged unsymmetrical as shown in the fig.(a)



The middle conductor will have more inductance because of more flux linkages from nearby conductors. As a consequence, less current flows through the middle SCR as compared to outer two SCRs. This unequal distribution can be avoided by mounting the SCRs symmetrically on the heat sink as shown in fig.(b)

(iv) **Uniform current sharing:**

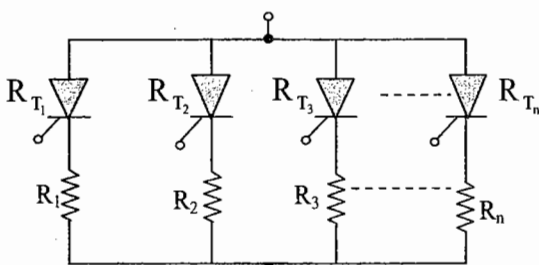
Current sharing in parallel connected SCRs can be made uniform by connecting a suitable external resistance in series with each SCR.

Where as $R_{T1}, R_{T2}, R_{T3}, \dots, R_{Tn}$ are Dynamic resistances of SCRs.

$R_1, R_2, R_3, \dots, R_n$ are external resistances.

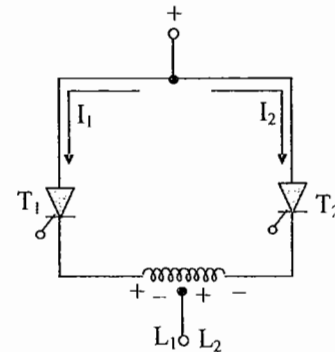
The values of $R_1, R_2, R_3, \dots, R_n$ are chosen such that

$$R_{T1} + R_1 = R_{T2} + R_2 = R_{T3} + R_3 = \dots = R_{Tn} + R_n.$$



(v) **Mid Point Reactor:**

In ac circuits, current distribution can be made more uniform by the magnetic coupling of the parallel paths. If currents I_1 and I_2 are unequal, say $I_1 > I_2$, then resultant flux linkages are not zero. These flux linkages induce mmf's in L_1 and L_2 . The reactor is arranged to completely balance the SCRs.

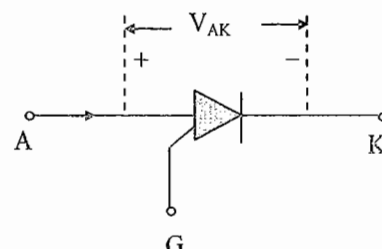


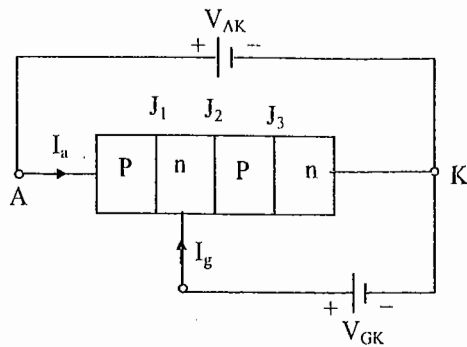
Other Members of Thyristor Family:

1. PUT (programmable uni-junction transistor)
2. SUS (silicon unilateral switch)
3. SCS (silicon controlled switch)
4. LASCR (Light Activated SCR)
5. SITHs (static induction thyristors)
6. The DIAC (Bidirectional Diode)
7. TRIAC (Anti parallel SCR'S)
8. ASCR (Asymmetrical SCR)
9. RCT (Reverse conducting thyristor)
10. G.T.O (Gate turn off thyristor)
11. MOSFET controlled thyristor (MCT)
12. FIELD controlled thyristor (FCT)

1. PUT (Programmable Uni Junction Transistor)

It is a P – N – P – N device like an SCR. But the major difference is that gate is connected to n-type material near the anode as shown in fig.

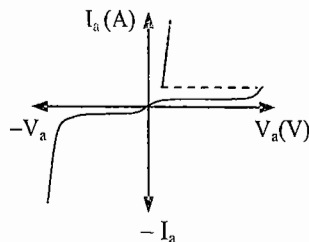




PUT is mainly used in time delay, logic and SCR triggering circuits.

It's largest rating is about 200V and 1A.

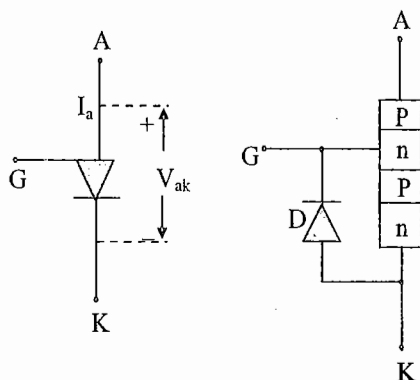
I-V characteristics of a PUT are.



Gate is always biased positive with respect to cathode.

2. SUS (Silicon Unilateral switch)

A SUS is similar to a PUT but with an inbuilt low-voltage avalanche diode between gate and cathode as shown in figure below.



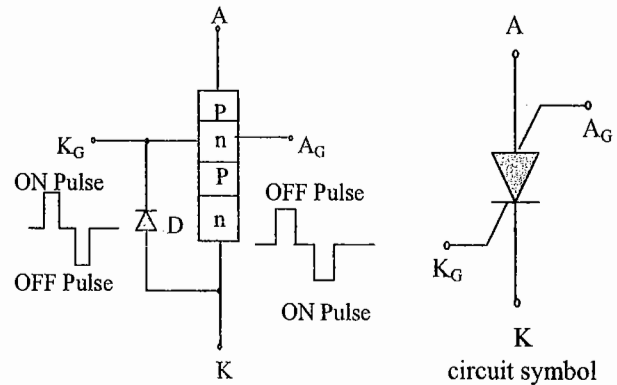
Because of presence of diode (D), SUS turns on for a fixed anode to cathode voltage. Its ratings are about 20V and 0.5A.

3. SCS (Silicon Controlled Switch):

SCS is a tetrode i.e. four electrode thyristor. It has two gates, one anode gate (A_G) like a PUT and another cathode gate (K_G) like an SCR.

SCS is a four layer, four terminal P – N – P – N device; with anode A cathode K, anode gate A_G and cathode gate K_G .

SCS can be turned on by either gate.

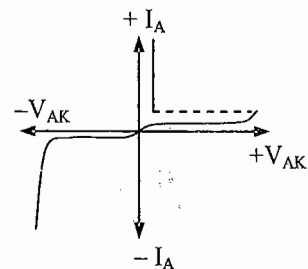


I – V characteristics

Anode gate is given negative pulse to turn ON

K_G is given positive pulse to turn ON.

Ratings SCS are 100V & 200 mA.



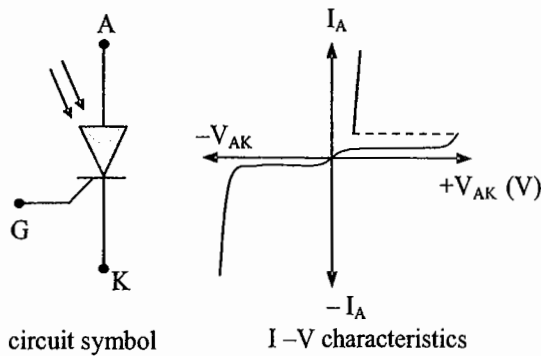
I – V characteristics of LASCR

4. LASCR (Light Activated SCR)

Light Activated SCR. It is turned on by throwing a pulse of light on the silicon wafer of thyristor.

If the intensity of light exceeds a certain value, excess electro hole pairs are generated due to radiation and forward – biased thyristor gets turned ON.

The rating of LASCR are upto 6kV and 3.5 kA with ON state voltage drop of 2V and with light – triggering requirement of 5mW.



5. SITHs (Static Induction Thyristor).

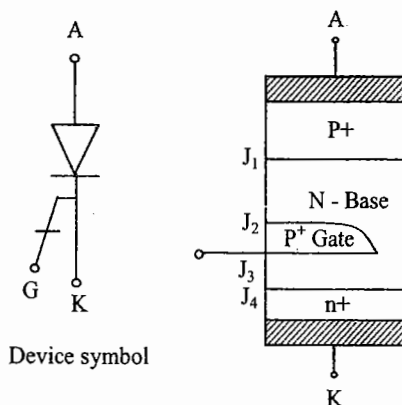
The SITH is also known as field controlled Diode (FCD).

The ratings of the device are $V = 2500V$ and $I = 500A$.

Turn-on of SITHs

A SITH is normally turned on by applying a positive gate voltage with respect to cathode. It's the gate cathode PN diode turns on and injects electrons from the N^+ cathode region into the base region between the P^+ gate and N^+ cathode, and into the channel.

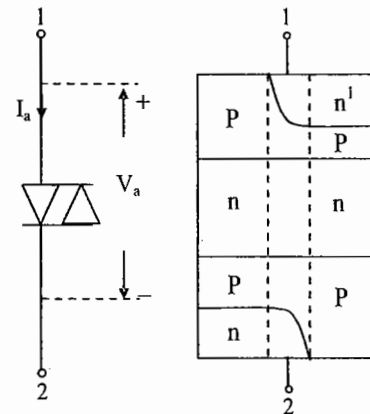
The positive gate voltage reduces the potential barrier in the channel, which gradually becomes conductive. When electrons reach the junction J_1 the P^+ anode begins to inject holes into the base, providing the base current of transistor Q_2 . As the base current increases, J_2 is eventually forward biased. The device is then fully turned on.



Turns off: An SITH is normally turned off by applying a negative voltage with respect to the cathode. If a sufficiently negative voltage is applied to the gate, a depletion layer forms around

P^+ gate, and eventually the depletion layer fully cuts off the channel.

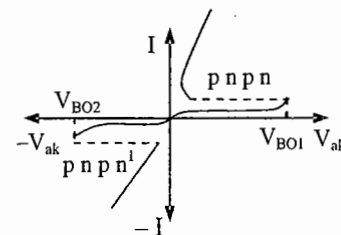
6. THE DIAC (Bidirectional Thyristor Diode)



If voltage V_{12} , with terminal 1 positive with respect to terminal 2, exceeds break over voltage V_{BO1} , then structure $P - N - P - N$ conducts.

In case terminal 2 is positive with respect to terminal 1 and when V_{21} exceeds break over voltage V_{BO2} structure $P - N - P - N$ conducts.

I - V characteristics of a DIAC



The term DIAC is obtained from capital letters, Diode that can work on AC.

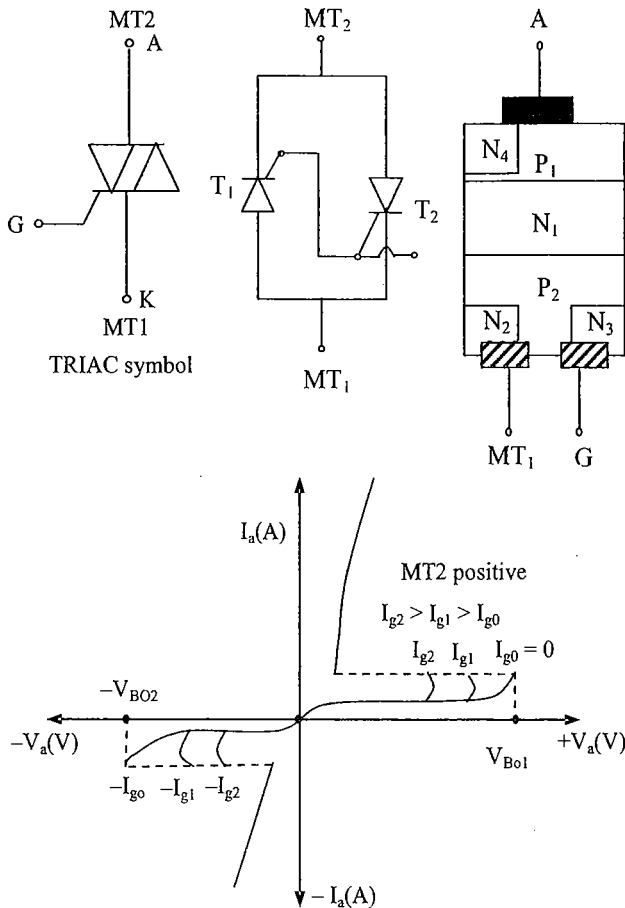
A diac is some times called a gate less TRIAC.

7. TRIAC:

A TRIAC is a bidirectional thyristor with three terminals. It is used extensively for the control of power in ac circuits.

TRIAC is the word derived by combining the capital letters from the word TRIode and AC.

Its three terminals are usually designated as MT_1 (Main terminal 1), MT_2 and the gate.

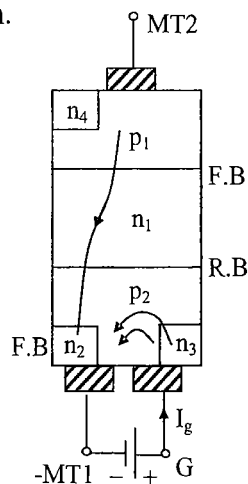


The triac can however, be turned on in each half cycle of the applied voltage by applying a positive voltage to the gate with respect to terminal MT₁.

The turn on process of a triac can be explained as under:

(i) MT₂ positive and gate current is also positive.

When gate current has injected sufficient charge into P₂ layer, reverse biased junction N₁ P₂ breaks down.



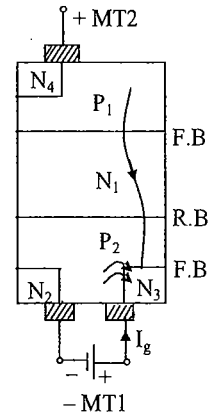
As a result, triac starts conducting through P₁, N₁ P₂, N₂ layers.

(ii) MT₂ is positive but gate is negative:

Gate current flows through P₂ N₃ junction.

Reverse biased junction N₁ P₂ is forward biased by injecting sufficient charge into P₂.

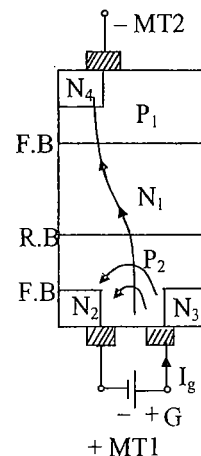
As a result triac starts conducting through P₁ N₁ P₂ N₃.



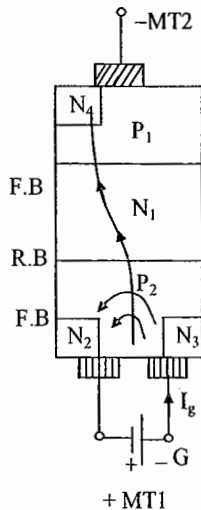
(iii) MT₂ is negative but gate current is positive:

Gate current I_g forward biases P₂ N₂ junction. Layer N₂ injects electrons into P₂ layer. As a result reverse biased junction N₁ P₁ breaks down.

The path P₂ N₁ P₁ N₄ is completely turned ON.



(iv) BOTH MT_2 AND GATE CURRENT ARE NEGATIVE:



The gate current I_g flows from P_2 to N_3 .

Reverse biased junction $N_1 P_1$ is broke and finally the structure $P_2 N_1 P_1 N_4$.

8. ASCR (Asymmetrical SCR)

An asymmetrical thyristor, is specially fabricated to have limited reverse voltage capability.

This permits a reduction in turn ON, turn off time and on state voltage drop in ASCR.

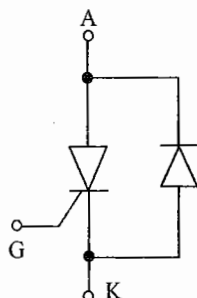
Fast turn off ASCR minimize the size, weight and cost of commutating components and permit high frequency operation (20 kHz or more) with improved efficiency.

9. REVERSE CONDUCTING THYRISTOR (RCT):

It may be consider as a thyristor with a built in antiparallel diode.

This construction reduces to zero the reverse blocking capability of RCT.

RCTs with 2000V and 500A rating are available.



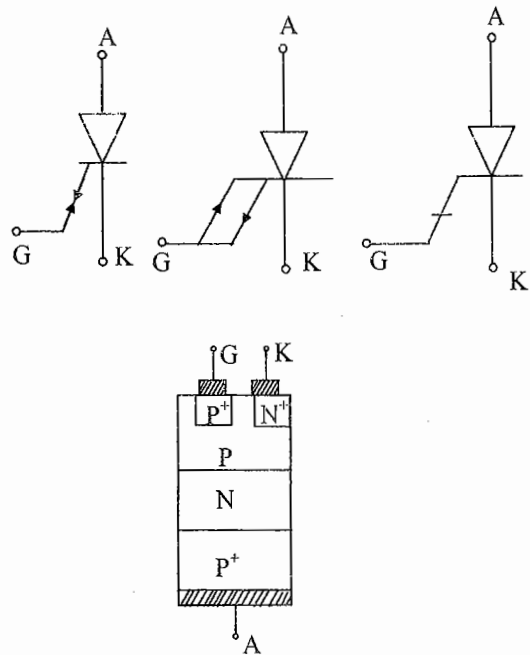
10 GTO (GATE TURN OFF THYRISTOR)

A GTO is a more versatile power semiconductor device.

It is like a conventional thyristor but with added features in it.

A GTO can be easily turned off by a negative gate pulse of appropriate amplitude. Thus a GTO is a device that can be turned – on by a positive gate current and turned – off by a negative gate current at its gate cathode terminals.

Self – turn off capability of GTO makes it the most suitable device for inverter and chopper applications.



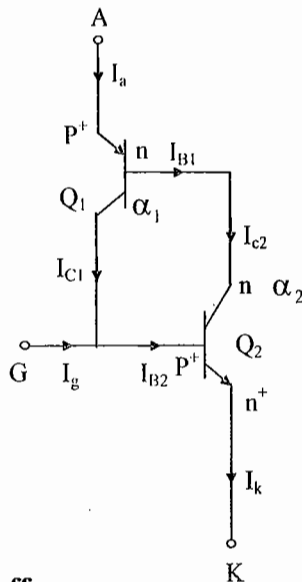
The four layers of GTO have different doping levels indicate by $P^+ n P^+ n^+$.

Transistor Q_1 is $P^+ NP^+$ type and Q_2 is NP^+N^+ . Emitter of Q_1 as anode and Emitter of Q_2 as cathode.

Turns on process:

A GTO is turned on by applying a positive gate current I_g in the direction shown.

Current gains α_1 and α_2 begins to rise and when $\alpha_1 + \alpha_2 = 1$, saturation level is reached and GTO is turned ON.



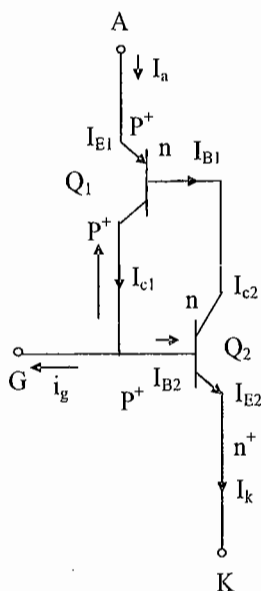
Turn off process:

For initiating the turn off process, Q_2 must be brought out of saturation.

The gate current I_g^1 is given in opposite side.

So when negative gate current I_g^1 flows between gate cathode terminals.

Net gate current is reversed & excess carriers are drawn from base P^+ region of Q_2 and collector current I_{C1} of Q_1 is diverted into the external gate circuit. This removes base drive of transistor Q_2 this further removes base current I_{B1} of transistor Q_1 and the GTO is eventually turned off.



Workedout Examples

12. The one-cycle surge current rating of a thyristor used in a 50 Hz circuit is 2000 A. Its $\frac{1}{4}$ cycle surge current rating will be

A. The blank is

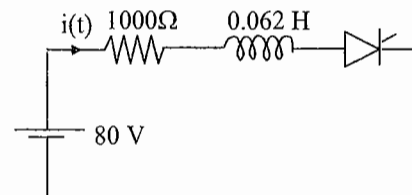
- (a) $2000\sqrt{2}$ (b) 500
(c) 8000 (d) $500\sqrt{2}$

Sol: Sub-cycle surge current rating $I_{sb} = I \times \sqrt{\frac{T}{t}}$

Where, for 50 Hz, T = time for one half cycle = 10 m sec; and t = time for one quarter cycle = 5 m sec.

$$I_{sb} = 2000 \sqrt{\frac{10 \times 10^{-3}}{5 \times 10^{-3}}} = 2000\sqrt{2}$$

13. The latching current of the thyristor shown in figure is 5 mA. The minimum duration for which gate current should be maintained is _____.



Ans: 4 μ s

Sol: Let the thyristor be turned on at $t = 0$.

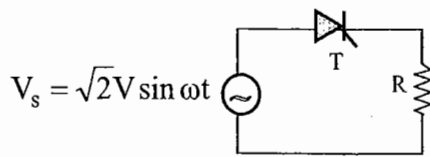
$$\text{For } t \geq 0^+ \quad i(t) = \frac{80}{1000} \left(1 - e^{\frac{-1000t}{0.062}} \right)$$

This current should reach the value 5 mA for the SCR to latch on.

$$\therefore \frac{80}{1000} \left(1 - e^{\frac{-1000t}{0.062}} \right) = 5 \times 10^{-3}$$

The unknown t can be found from this equation and works out to be 4 μ sec.

14. In the given circuit a thyristor converter is feeding a resistor R . The power consumed by R in the absence of SCR is P . In the presence of SCR, the power consumed by R at $\alpha = 90^\circ$ would be reduced by a factor of



- (a) $\frac{1}{2}$ (b) $\frac{1}{4}$
(c) $\frac{1}{6}$ (d) $\frac{1}{12}$

Sol: In the absence of SCR $P_1 = \frac{V_{or}^2}{R} = \frac{V^2}{R}$

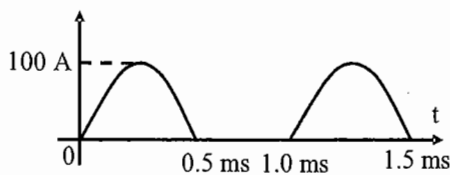
In the presence of SCR $\Rightarrow V_{or} =$

$$\frac{V_m}{\sqrt{2\pi}} \left[(\pi - \alpha) + \frac{1}{2} \sin 2\alpha \right]^{\frac{1}{2}}$$

$$\text{For } \alpha = 90^\circ \Rightarrow V_{or}^2 = \frac{V^2}{3} \Rightarrow P_2 = \frac{V^2}{4R}$$

$$\frac{P_2}{P_1} = \frac{1}{4}$$

15. A thyristor is used in an application carrying half sinusoidal current of period of 1 msec and a peak of 100 A as shown in fig. The thyristor may modeled during conduction to have a constant voltage drop of 1.1V and a dynamic resistance of 8 m Ω . Evaluate the average conduction loss in the device



- (a) 35 W (b) 0.4 W
(c) 55 W (d) 20 W

Sol: Average conduction loss = $I_{av} \times V_T + I_{rms}^2 R$
= 55 W

16. An SCR has half cycle surge current rating of 3000 A for 50 Hz supply. One cycle surge current rating will be
(a) 1500 A (b) 2121.32 A
(c) 4242.64 A (d) 6000 A

17. Two thyristors are connected in parallel to share a load current of $I_L = 600$ A. The on state voltage drop of a one thyristor is $V_{T1} = 1.0$ V at 300 A and that of other thyristor is $V_{T2} = 1.5$ V at 300 A. Determine the values of series resistances to force current sharing with 10% difference. Take voltage $V = 2.5$ V.

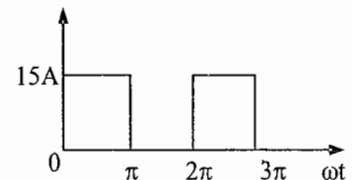
Sol: (Voltage across $T_1 + 330R$) = (voltage across $T_2 + 270R$)

$$1 + 330R = 1.5 + 270R$$

$$R = \frac{1}{120} \Omega$$

18. A MOSFET rated for 20 A carries a periodic current as shown in fig. The ON state resistance of the MOSFET is 0.12 Ω . The average ON state loss in the MOSFET is

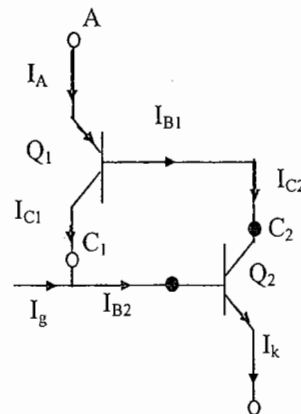
- (a) 33.8 W
(b) 13.5 W
(c) 7.5 W
(d) 3.8 W



Sol: The average ON state loss

$$= \frac{1}{2\pi} \int_0^\pi (15)^2 \times 0.12 dt = 13.5 \text{ W}$$

19. In figure, Q_1 and Q_2 have common base current gain values of α_1 and α_2 respectively. (at the emitter currents involved). $I_a =$



- (a) $\frac{\alpha_2 I_g}{1 - (\alpha_1 + \alpha_2)}$ (b) $\frac{(\alpha_1 + \alpha_2) I_{gt}}{\alpha_2}$
(c) $\frac{\alpha_2 I_g}{1 + (\alpha_1 + \alpha_2)}$ (d) $\frac{\alpha_1 I_g}{1 - (\alpha_1 + \alpha_2)}$



Sol: From the figure, $I_{C1} = \alpha_1 I_a$ and $I_{C2} = \alpha_2 I_k$
(I_a is the emitter current of Q_1 and I_k the emitter current of Q_2)

By KCL;

$$1. I_a = I_{C1} + I_{C2} = \alpha_1 I_a + \alpha_2 I_k$$

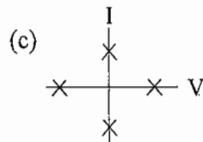
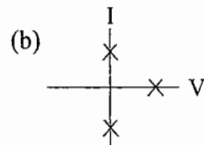
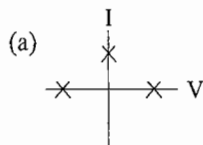
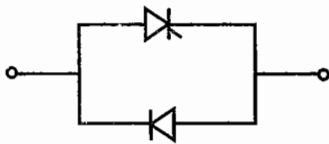
$$2. I_{C1} = I_{C2} + I_g = I_k = I_a + I_g$$

$$\therefore I_a = \alpha_1 I_a + \alpha_2 I_a + \alpha_2 I_g$$

$$I_a [1 - (\alpha_1 + \alpha_2)] = \alpha_2 I_g \text{ from which}$$

$$I_a = \frac{\alpha_2 I_g}{1 - (\alpha_1 + \alpha_2)}$$

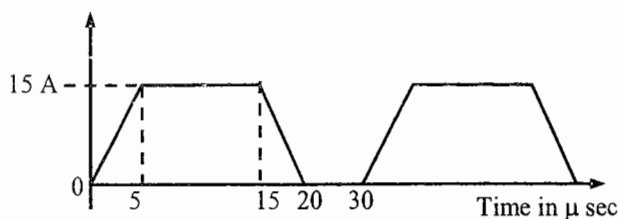
20. A power electronic switch is realized by SCR and diode as shown below. Identify the static operating points on V-I plane



(d) Invalid switch combination

Ans: (b)

21. The wave form in the following figure shows the periodic current through a power-switching device in a switching converter application.



A BJT with a device drop of 1.5 V and MOSFET with an ON state resistance of 200 mΩ are considered for this application. The conduction loss in BJT and MOSFET are respectively

- (a) 11.25 & 10 W (b) 11.25 W & 20 W
(c) 10 W & 11.25 W (d) 20 W & 11.25 W

Sol: Conduction loss in BJT = $I_{av} \times V_{sw}$
= 11.25 W

$$\text{Conduction loss MOSFET} = I_{rms}^2 \times R_{ds} = 20 \text{ W}$$

22. A BJT with current gain $\beta = 50$ has load resistance $R_L = 10\Omega$, d.c. supply voltage $V_{cc} = 120\text{V}$ and input voltage to base circuit $V_B = 10\text{V}$. For $V_{CES} = 1.2\text{V}$ and $V_{BES} = 1.6\text{V}$ calculate
(a) the value of R_B for operation in the saturated state.
(b) the value of R_B for over drive factor = 6
(c) Forced current gain
(d) Power loss in transistor.

Sol: $\beta = 50$, $R_C = 10\Omega$, $V_{cc} = 120\text{V}$

$V_B = 10\text{V}$, $V_{CES} = 1.2\text{V}$ and $V_{BES} = 1.6\text{V}$

$$(a) I_{cs} = \frac{V_{CC} - V_{CES}}{R_C} = \frac{120 - 1.2}{10} = 11.88 \text{ A}$$

$$I_{BS} = \frac{11.88}{50} = 0.2376$$

$$R_B = \frac{V_B - V_{BES}}{I_{BS}} = \frac{10 - 1.6}{0.2376} = 35.35\Omega$$

(b) ODF = 6

$$\text{ODF} = \frac{I_B}{I_{BS}}$$

$$I_B = 6 \times 0.2376 = 1.4256$$

$$R_B = \frac{V_B - V_{BES}}{I_B} = \frac{10 - 1.6}{1.4256} = 5.89\Omega$$

(c) Forced current gain

$$\beta_f = \frac{I_{cs}}{I_B} = \frac{11.88}{1.4256} = 8.33$$

(d) Power loss $P_T = V_{BE} I_B + V_{CE} I_C = 14.63 \text{ W}$

23. The primary of a pulse transformer is connected in series with a transistor and a current limiting resistor R_L . The data for the triggering circuit is as under. $R_L = 500\Omega$, gate cathode resistance = 200Ω primary to secondary turns ratio = $\frac{1}{2}$, voltage required to trigger the SCR = 3V. The voltage applied to the circuit consisting of transformer primary is

- (a) 16.5 V (b) 24.75 V
(c) 33 V (d) 49.5 V



Sol: Given data,

$$R_L = 500\Omega \quad R_g = 200\Omega \quad \frac{N_1}{N_2} = 1/2$$

$$V_{SCR} = 3V$$

$$R_1 = \left(\frac{N_1}{N_2} \right)^2 R_g = \frac{1}{4} \times 200 = 50\Omega$$

The SCR voltage on primary side is $= 3/2V$

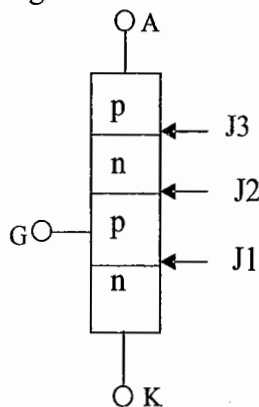
$$e_0 = \frac{V_B R_1}{R_1 + R_L}$$

$$\frac{3}{2} = \frac{V_B \times 50}{550} \Rightarrow V_B = 16.5V$$

Class Room Practice Questions

LEVEL-1 Questions

01. Figure shows a thyristor with the standard terminations of anode (A), cathode (K), gate (G) and the different junctions named J1, J2 and J3. When the thyristor is turned ON and conducting



- (a) J1 and J2 are forward biased and J3 is reverse biased
 (b) J1 and J3 are forward biased and J2 is reverse biased
 (c) J1 is forward biased and J2 and J3 are reverse biased
 (d) J1, J2 and J3 are all forward biased
02. The snubber circuit is used in thyristor circuits for
 (a) triggering (b) dv/dt protection
 (c) di/dt protection (d) phase shifting
03. It is preferable to use a train of pulse of high frequency for gate triggering of SCR in order to reduce
 (a) dv/dt problem
 (b) di/dt problem
 (c) the size of the pulse transformer
 (d) the complexity of the firing circuit
04. Turn – on & turn – off times of transistor depend on
 (a) static characteristic
 (b) junction capacitances
 (c) current gain
 (d) None of the above
05. Which one of the following statements is correct? The turn off times of converter grade SCRs are normally in the range of
 (a) 1 to 2 microseconds
 (b) 50 to 200 microseconds
 (c) 500 to 1000 microseconds
 (d) 1 to 2 milliseconds
06. Which one of the following statements is correct? In order to get best results per unit cost, the heat sinks on which the thyristors are mounted, are made of
 (a) aluminium (b) copper
 (c) nickel (d) stainless steel
07. Power electronic device with poor turn off gain is
 (a) A symmetrical thyristor
 (b) A conventional thyristor
 (c) Power bipolar junction transistor
 (d) Gate turn – off thyristor
08. A resistor connected across the gate and cathode of an SCR
 (a) increase $\frac{dv}{dt}$ rating of SCR
 (b) increases holding current of SCR
 (c) increase noise immunity of SCR
 (d) increase turn – off time of SCR
09. A thysistor equivalent of a thyatron tube is a
 (a) SCR (b) UJT
 (c) Diac (d) Triac
(IES-EE-92)
10. When a thyristor is negatively biased
 (a) All the three junctions are negatively biased
 (b) Outer junctions are positively biased and the inner junction is negatively biased.
 (c) Outer junctions are negatively biased and the inner junction is positively biased
 (d) The junction near the anode is negatively biased and the one near the cathode is positively biased
(IES-EE-97)



11. Consider the following semiconductor devices:

1. Triac
2. Thyristor
3. Amplifying gate thyristor.

The correct sequence of these devices in increasing order of their di/dt capabilities is (IES-EE-97)

- (a) 1, 3, 2 (b) 1, 2, 3
(c) 3, 1, 2 (d) 3, 2, 1

12. If the amplitude of the gate pulse to thyristor is increased, then (IES-EE-98)

- (a) both delay time and rise time would increase.
(b) the delay time would increase but the rise time would decrease
(c) the delay time would decrease but the rise time would increase
(d) the delay time would decrease while the rise time remains unaffected

13. In order to obtain static voltage equalization in series – connected SCRs, Connections are made of (IES-EE-98 & 99)

- (a) One resistor across the string
(b) resistors of different value across each SCR
(c) resistors of the same value across each SCR
(d) one resistor in series with the string

14. The correct sequence of the given devices in the decreasing order of their speeds of operation is (IES-EE-99)

- (a) Power BJT power MOSFET, IGBT, SCR
(b) IGBT, power MOSFET, power BJT, SCR
(c) SCR, power BJT, IGBT, MOSFET
(d) MOSFET, IGBT, power BJT, SCR

15. During the turn - off process of a thyristor, the current flow, does not stop at the instant when the current reaches zeros but continues to flow to a peak value in the reverse direction. This is due to (IES-EE-00)

- (a) commutation failure

(b) hole storage effect

(c) presence of reverse voltage across the thyristor

(d) protective inductance in series with the thyristor

16. Consider the following statements:

1. BJT has lower power losses than MOSFETs.
2. MOSFETs have lower power losses than IGBTs
3. SCRs have lower power losses than MOSFETs and IGBTs

Which of these statements are correct?

(IES-EE-00)

- (a) 1, 2 and 3 (b) 1 and 2
(c) 2 and 3 (d) 1 and 3

17. When compared to those of a symmetrical thyristor, the turn-off time and reverse blocking voltage of an asymmetrical thyristor are respectively (IES-EE-00)

- (a) large and large (b) large and small
(c) small and large (d) small and small

18. Triac cannot be used in (IES-EE-01)

- (a) ac voltage regulators
(b) cycloconverters
(c) solid state type of switch
(d) inverter

19. The most suitable device for high frequency inversion in SMPS is (IES-EE-01)

- (a) BJT (b) IGBT
(c) MOSFET (d) GTO

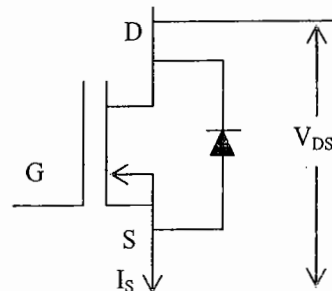
20. When cathode of a thyristor is made more positive than its anode (IES-EE-02)

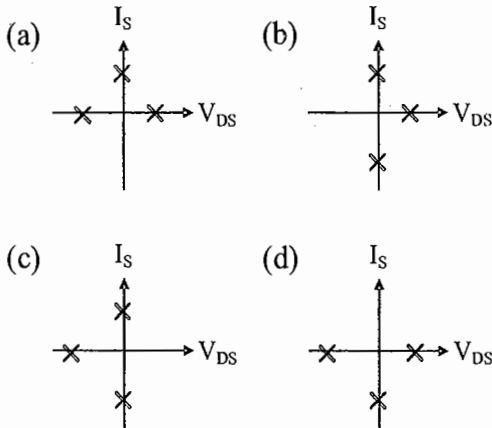
- (a) all the junctions are reverse biased
(b) outer junctions are reverse biased and central one is forward biased
(c) outer junctions are forward biased and central one is reverse biased
(d) all the junctions are forward biased

21. The sharing of the voltages between thyristors operating in series is influenced by the (IES-EE-02)

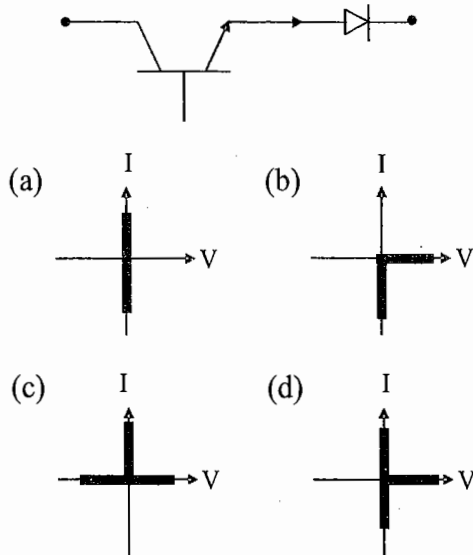


- (a) di/dt capabilities
(b) dv/dt capabilities
(c) junction temperatures
(d) static V-I characteristics and leakage currents
22. A gate turn off (GTO) thyristor has capacity to **(IES-EE-05)**
(a) Amplify the gate – current
(b) Turn – off when positive current pulse is given at the gate
(c) Turn – off when a gate – pulse is given at the gate even though it is reverse biased
(d) Turn – off when a negative current pulse is given at the gate
23. Snubber circuits are used to protect thyristor from which of the following? **(IES-EE-07)**
(a) High $\frac{di}{dt}$ and low $\frac{dv}{dt}$
(b) High $\frac{dv}{dt}$ and low $\frac{di}{dt}$
(c) Low $\frac{dv}{dt}$ and low $\frac{di}{dt}$
(d) High $\frac{dv}{dt}$ and high $\frac{di}{dt}$
24. A modern power semiconductor device that combines the characteristics of BJT and MOSFET is **(IES-EE-08)**
(a) GTO (b) FCT
(c) IGBT (d) MCT
25. The conduction loss versus device current characteristic of a power MOSFET is best approximated by **(GATE-05)**
(a) a parabola
(b) a straight line
(c) a rectangular hyperbola
(d) an exponentially decaying function
26. The following is generally true RC snubber is used to protect the SCR **(JTO-01)**
(a) against device over voltage
(b) against device over current.
(c) against forward dv/dt current
(d) against di/dt
27. The following is generally true. A series inductor is used to protect the SCR **(JTO-02)**
(a) against device over voltage
(b) against device over current
(c) against forward dv/dt
(d) against di/dt
28. Which one of the following Statements is **True** for an ‘ideal’ power diode? **(JTO-03)**
(a) Forward voltage drop is zero and reverse saturation current is non zero
(b) Reverse recovery time is non-zero and reverse saturation current is zero
(c) Forward voltage drop is Zero and reverse recovery time is zero
(d) Forward voltage is non zero and reverse recovery time is zero
29. Which one of the following is NOT True? **(JTO-09)**
(a) For SCRs to be in the conduction state, the forward anode current must be greater than the latching current
(b) For SCRs to be in the forward blocking state, the forward anode current must be lower than the holding current
(c) When SCRs are in the conduction state, they can be turned off by applying suitable gate pulses
(d) When avalanche breakdown takes place, SCRs enter into the conduction state
30. Figure shows a MOSFET with an integral body diode. It is employed as a power switching device in the ON and OFF states through as a power switching device in the ON and OFF states through appropriate control. The ON and OFF states of the switch are given on the $V_{DS} - I_S$ plane by **(GATE- 03)**





31. Figure shows a composite switch consisting of a power transistor (BJT) in series with a diode. Assume that the transistor switch and the diode are ideal; the I-V characteristic of the composite switch is **(GATE-10)**

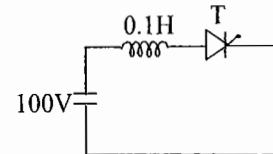


32. Which one of the following is used as the main switching element in a switched mode power supply operating in 20 kHz to 100 kHz range? **(IES - 04)**
- (a) Thyristor (b) MOSFET
(c) Triac (d) UJT
33. A power diode is in the forward conduction mode and the forward current is now decreased. The reverse recovery time of the diode is t_r and the rate of fall of the diode current is di/dt . What is the stored charge? **(IES - 04)**

- (a) $(di/dt) t_r$ (b) $1/2 (di/dt) t_r$
(c) $1/2 (di/dt) t_r$ (d) $1/2 (di/dt) t_r$

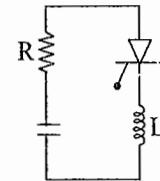
LEVEL-2 Questions

01. The latching current of SCR in the given circuit is 4 mA. The minimum width of the gate pulse required to properly turn on the thyristor is



- (a) 6 μ s
(b) 4 μ s
(c) 2 μ s
(d) 1 μ s

02. The latching current of a SCR is 12 mA. Its holding current will be

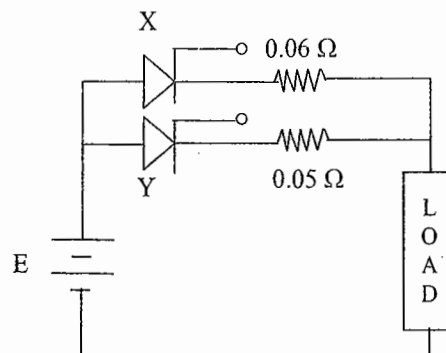


- (a) 4 mA
(b) 12 mA
(c) 50 mA
(d) 8 mA

03. An SCR with heat sink arrangement has ambient temperature 50°C and junction temperature 100°C . Average power dissipated is 50 W. If junction temperature is reduced to 70°C , percentage reduction in thermal resistance is

- (a) 60% (b) 40%
(c) 50% (d) 30%

04. The figure shows two thyristors each rated 500A (continuous) sharing a load current. Current through thyristors Y is 120 A. The current through thyristor X will be nearly ---A.





05. The anode current through a conducting SCR is 10 A. If its gate current is made one – fourth, then what will be the anode current?

(IES-EE-09)

- (a) 0 A (b) 5 A
(c) 10 A (d) 20 A

06. The forward voltage of a power diode is $V_D = 1.0$ V at the diode current $I_D = 200$ A. Assuming the emission coefficient $\eta = 2$ and the thermal voltage $V_t = 25.7$ mV the reverse Saturation current I_s (Approximate) is.

(JTO-2009)

- (a) $71.08 \mu\text{A}$ (b) $51.20 \mu\text{A}$
(c) $21.09 \mu\text{A}$ (d) $41.09 \mu\text{A}$

07. For an SCR, the gate cathode characteristic has a straight line slope of 140. For trigger source voltage of 20V and allowable gate power dissipation of 0.5 Watts, what is the gate source resistance? (IES-EE-09)

- (a) 200Ω (b) 255Ω
(c) 195Ω (d) 185Ω

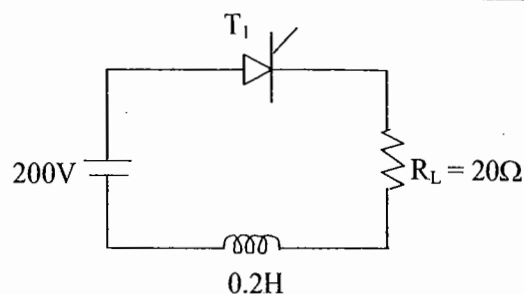
08. An SCR is rated for 650 V PIV. What is the voltage for which the device can be operated if the voltage safety factor is 2? (IES-EE-09)

- (a) 325V rms (b) 230V rms
(c) 459V rms (d) 650V rms

09. A dc source of 100 volts supplies a purely inductive load of 0.1 H; the controller is an SCR in series with source and load. If the specified latching current is 100 mA, then the minimum width of the gating pulse to ensure turn – on of SCR would be (IES-EE-12)

- (a) $10 \mu\text{s}$ (b) $50 \mu\text{s}$
(c) $100 \mu\text{s}$ (d) $1 \mu\text{s}$

10. The latching current of T_1 is 1 mA. The minimum width of gate pulse required to turn on SCR is (AP GENCO-12)



- (a) $2 \mu\text{sec}$ (b) $1 \mu\text{sec}$
(c) $0.5 \mu\text{sec}$ (d) $1.5 \mu\text{sec}$

11. For an SCR, the gate cathode characteristic has a straight line slope of 100. For trigger source voltage of 15 V and allowable gate power dissipation of 1W, compute the gate source resistance.
12. A thyristor is triggered by a train of pulses of frequency 4 kHz and of duty cycle 0.2. Calculate the pulse width. In case average power dissipation is 1W, find the maximum allowable gate power of the drive.
13. The junction capacitance of a thyristor can be assumed to be independent of off state voltage. The limiting value of charging current to turn on thyristors is $12 \mu\text{A}$. If the critical value of dv/dt is $800 \text{ V}/\mu\text{s}$, determine the junction capacitance.
14. For a power diode, $t_{rr} = 3.9 \mu\text{sec}$ and rate of diode current decay is $50 \text{ A}/\mu\text{sec}$ for a softness factor of 0.3. Find peak inverse current and stored charge.



Solutions

LEVEL-1

01. **Ans:** (d)

02. **Ans:** (b) & (c)

Sol: Snubber circuit is a series combination of resistor R and a capacitor C, Placed across the thyristor, for protection of SCR against $\frac{di}{dt}$ turn on.

03. **Ans:** (c)

Sol: For High frequencies the size and weight of the transformer are less, due to this area of core decreases.

04. **Ans:** (b)

Sol: Any p-n junction offer capacitance. Due to these capacitances voltage variation gets delayed, it is the origin for turn on and turn off times.

05. **Ans:** (b)

Sol: Slow turnoff time (50 - 100 μ s) SCR's are called converter grade SCR's and fast turn off time (3 - 50 μ sec) are called Inverter grade SCR's

06. **Ans:** (a)

Sol: Heat sinks are made using metals having high thermal conductivity. Aluminium is the most commonly used metal, due to economy and reasonable thermal conductivity.

07. **Ans:** (d)

Sol: In a power device,

$$\text{current gain} = \frac{\text{Anode current}}{\text{Gate current}}$$

In GTO higher value of gate current is required to make turn off. Hence turn off gain would be less.

08. **Ans:** (c)

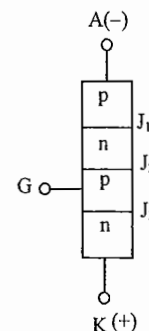
Sol: A resistor connected across gate and cathode of an SCR by pass noise signal and avoid mal triggering due to noise signal.

09. **Ans:** (a)

Sol: The metal tank rectifier, grid-controlled vacuum-tube rectifier, ignition, phanotron and thyatron were introduced one after another for power control until the 1950's. After that four-layer device called thyristors were developed in 1958. Thyatrons are equivalent to SCR's

10. **Ans:** (c)

Sol: When SCR is positively biased (i.e. A is +ve, K is - ve) then J_1 & J_3 are positively biased and J_2 is negatively biased. When SCR is negatively biased (i.e A is - ve, K is + ve) then J_1 & J_3 are reverse biased and J_2 is forward Biased.



11. **Ans:** (c)

Sol: Thyristor has good $\frac{di}{dt}$ capability by using series inductor. Triac has same Characteristics as SCR. But triac conducts in reverse biased mode with suitable gate pulsc.

12. **Ans:** (d)

Sol: Delay time can be decreased by applying high gate current and more forward voltage between A & K, but rise time can be decreased if high and steep current pulses are applied to gate.

13. **Ans:** (c)

Sol: In series connected SCR's uniform voltage distribution in steady state can be achieved



by connecting a suitable resistance across each SCR such that each parallel combination has the same resistance. This will require different value of resistance for each SCR which is a difficult proposition. A more practical way of obtaining a reasonably uniform voltage distribution during steady state working of series-connected SCR's is to connect the same value of shunt resistance 'R' across each SCR.

14. **Ans:** (d)

15. **Ans:** (b)

Sol: During time t_q all the excess carriers from the four layers of SCR must be removed. This removal of excess carriers consists of sweeping out of holes from outer p - layer and electrons from outer n - layer.

16. **Ans:** (d)

Sol: 1. MOSFET has lower switching losses and has high conduction losses but BJT has high switching losses and has less conduction losses.

2. IGBT has lower switching losses than BJT. Therefore the power loss in increasing order is SCR, BJT, IGBT and MOSFET.

17. **Ans:** (d)

Sol: Turn off time and reverse blocking voltage both are small in asymmetrical thyristor and also ON state voltage drop is less.

In ASCR reverse blocking voltage capability 20 to 30V and forward blocking voltage capability 400 to 2000 V.

ASCR is used in high frequency operations of less turn off time.

18. **Ans:** (c)

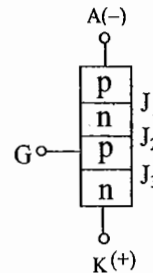
Sol: Triac is a bidirectional device. We can't use this as a solid state type of switch because switch is a unidirectional.

19. **Ans:** (c)

Sol: MOSFET has low switching power loss and high switching frequency.

20. **Ans:** (b)

Sol: J_1, J_3 Outer Junctions are forward biased.
 J_2 Inner Junction is reverse biased



21. **Ans:** (d)

Sol: The thyristors will have difference in static V-I characteristics, which in turn results into unequal leakage resistances and currents. The sharing of voltage across thyristors depends on leakage resistance and currents.

22. **Ans:** (d)

Sol: GTO can be turn off by applying negative gate current while turn on process is same as SCR.

23. **Ans:** (d)

Sol: Voltage snubber (R&C in series) used for high dv/dt. The resistance in snubber circuit limits discharge current of capacitor to a lower value, hence it also decreases di/dt.

24. **Ans:** (c)

Sol: IGBT has high input impedance as MOSFET and has low forward voltage drop as BJT

25. **Ans:** (c)

Sol: Conduction loss of MOSFET = $I_a^2 R_{ds(on)}$ loss
 V_s I_{ds} curve is rectangular hyperbola.

26. **Ans:** (c)

27. **Ans:** (d)

28. **Ans:** (c)

29. **Ans:** (c)

30. **Ans:** (b)

Sol: MOSFET with anti parallel diode consists of forward blocking state, forward conduction state and reverse conduction state.



31. **Ans:** (c)

32. **Ans:** (b)

Sol: MOSFET has lower switching losses and operated at higher frequencies

33. **Ans:** (c)

Sol: Assume reverse recovery characteristic is triangular then stored charge

$$Q = \frac{1}{2} I_{RM} t_r$$

peak inverse current

$$I_{RM} = \left(\frac{di}{dt} \right) t_r$$

$$\therefore Q = \frac{1}{2} \left(\frac{di}{dt} \right) t_r^2$$

Level-2 Key

01. (b) 02. (a) 03. (a) 04. 100A

05. (c) 06. (d) 07. (c) 08. (b)

09. (c) 10. (b) 11. 50Ω 12. 50 μsec,
5 W

13. 0.015 pF 14. 150 A,
292.5 μsec